

Eddy Current Separator

Design, Fabrication, and IoT-Integrated Real-Time Monitoring System

Engineering Documentation & Technical Specification

Abstract — This technical report documents the design, fabrication, and experimental evaluation of a modular, benchtop-scale Eddy Current Separator (ECS) equipped with an integrated real-time Internet of Things (IoT) monitoring system. Eddy current separation leverages alternating magnetic fields induced by a high-speed permanent magnet rotor to generate opposing Lorentz forces in conductive non-ferrous particles, deflecting them away from non-conductive and ferrous bulk streams. The prototype integrates an adjustable conveyor drive system, an internal insulated permanent magnet rotor assembly, and an embedded sensing subsystem utilizing an Arduino microcontroller, an HX711-amplified load cell for continuous material mass logging, and an ultrasonic sensor for bin level monitoring. Telemetry is streamed wirelessly via an HC-05 Bluetooth transceiver to a dedicated Android supervisory dashboard. Empirical tests performed on aluminum, copper, and brass test specimens validate separation deflections between 0.9 m and 1.7 m. A comprehensive breakdown of mechanical drafting, bill of materials, circuit topology, assembly procedures, operational bottlenecks, and resolutions is provided to support open reproducibility.

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1 Introduction

1.1 Background

Industrial waste segregation and resource recovery represent critical challenges in municipal and scrap management ecosystems worldwide. Projected global municipal solid waste generation is expected to rise by over 70% by 2050. Despite the urgent need for recycling, low recovery rates persist in developing regions due to financial and operational constraints. Recovering recyclable commodities—particularly high-value non-ferrous metals such as aluminum, copper, and brass—dramatically reduces the energy consumption and carbon footprint associated with primary metal smelting and raw ore extraction.

Automating the recovery process eliminates human exposure to sharp, contaminated, or hazardous waste fragments while boosting processing throughput. Effective separation at the source also curbs downstream landfill overflow and preserves critical metallic reserves.

1.2 Working Principle of Eddy Current Separation

An Eddy Current Separator (ECS) operates on the physical principles of Faraday’s Law of Electromagnetic Induction and the Lorentz force equation. When a time-varying magnetic field interacts with an electrically conductive non-ferrous material, eddy currents are induced within the body of the material:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (1)$$

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

where $\mathbf{B}$ is the magnetic flux density, $\mathbf{E}$ is the induced electric field, σ is the electrical conductivity, and $\mathbf{J}$ is the induced eddy current density.

These circulating currents generate an intrinsic opposing secondary magnetic field in accordance with Lenz’s Law. The interaction between the external magnetic field gradients and the induced eddy currents produces a net repulsive Lorentz force:

$$\mathbf{F} = \int_V (\mathbf{J} \times \mathbf{B}) dV \quad (3)$$

In an ECS, a high-speed rotor lined with alternating polarity permanent magnets rotates within a non-metallic shell at the discharge head of a conveyor belt. As a mixed waste stream approaches the discharge pulley, non-conductive materials (such as plastics, wood, and glass) fall freely along a natural parabolic gravitational trajectory. In contrast, conductive non-ferrous particles experience an intense repulsive trajectory kick that throws them forward over an adjustable splitter blade into an isolated collection vessel.

1.3 Contribution to the Metal Recycling Industry

Eddy current separators are ubiquitous in municipal waste recycling, electronic scrap (e-waste) processing, automobile shredder residue (ASR) recovery, and foundry sand cleanup. Commercial ECS systems routinely achieve non-ferrous metal recovery rates of up to 95%, substantially decreasing dependency on virgin mining operations, cutting landfill disposal surcharges, and improving the economic yield of recycling facilities.

2 Problem Definition

2.1 Problem Statement

“Inefficient waste management in developing industrial contexts is exacerbated by the absence of localized, modular non-ferrous metal separation machinery. By integrating

real-time telemetry and sensing frameworks into an automated separator, facility managers can monitor mass recovery rates, optimize operational parameters, and enhance scalability.”

2.2 Key Industrial Challenges

Conventional sorting facilities face several structural bottlenecks:

- 1. Loss of Recoverable Non-Ferrous Metals:** Manual or low-field sorting systems fail to capture small particles and non-magnetic conductive fragments, dumping valuable raw stock into landfills.
- 2. High Human Labor Dependency and Low Sorting Efficiency:** Manual hand-picking suffers from low throughput, worker injury risks from hazardous contaminants, and inconsistent purity yields.
- 3. Absence of Real-Time Operational Feedback:** Traditional scrap sorting machines operate as open-loop units without automated logging of collected weight, capacity status, or material throughput.

3 Literature Review

A review of recent academic literature and technical patents informed the physical architecture and telemetry systems of this project, as summarized in Table 1.

Table 1: Summary of Surveyed Literature and Project Relevance

Paper Title	Author / Year	Key Findings	Project Relevance
Effect of eddy currents on metal fine particles in high gradient magnetic separation using a superconducting magnet	H. Chen & O. Miura (2024)	Investigated dry high-gradient magnetic separation for fine metallic particles. Demonstrated that capture efficiency is strongly governed by particle dimensions, material conductivity, and rotor field dynamics.	Supplied baseline deflection criteria across aluminum, copper, and precious metals, underscoring the influence of rotor rotational velocity.
Separation of copper from aluminum in an eddy current separator product by dry density separation	M. Ali (2024)	Analyzed fluidized bed and pneumatic density separation methods to isolate Cu from Al post-ECS sorting. Evaluated efficiency against manual sorting and XRF validation.	Informed downstream collection architecture, showing that high-density non-ferrous scrap requires secondary density classification or increased field strength.

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Table 1 continued from previous page

Paper Title	Author / Year	Key Findings	Project Relevance
Experiment sorting of non-ferrous metals by new orientation of magnetic disk separator	K. Nadjib et al.	Modeled an alternative single-disk vertical ECS architecture using alternating permanent magnets. Quantified the relationship between disk speed, particle geometry, and ejection trajectory.	Offered structural insights into drum configurations, magnetic pole spacing, and speed ratios between drive pulleys and internal rotors.
Digitalization and intelligent robotics in value chain of circular economy oriented waste management	R. Sarc et al.	Explored Industry 4.0 paradigms (ReWaste 4.0) integrating sensory digitization, automated robotics, and real-time level/weight acquisition for supply chain optimization.	Guided the selection of an IoT sensory stack (load cell, ultrasonic sensor, Bluetooth telemetry) for live collection bin monitoring.

4 Project Objectives

4.1 Primary Objective

To design, engineer, and validate a fully functional benchtop prototype of an Eddy Current Separator capable of separating non-ferrous metallic fragments (Al, Cu, Brass) from non-conductive municipal/industrial waste.

4.2 Specific Technical Objectives

- **Mechanical Framework:** Fabricate a rigid, vibration-isolated aluminum extrusion chassis housing independent drive belts and a high-speed magnetic rotor.
- **Independent Speed Control:** Implement dual motor drives managed via a Variable Frequency Drive (VFD) and reduction gearing to optimize sorting efficiency.
- **Real-Time Mass Logging:** Integrate a 20 kg strain-gauge load cell and 24-bit HX711 ADC to quantify recovered scrap weight dynamically.
- **Bin Level Monitoring:** Integrate an ultrasonic time-of-flight sensor to detect container overflow thresholds.
- **Wireless Supervisory Interface:** Develop an Android dashboard application for real-time visualization of sorting metrics and automated bin-full alerts.

5 System Architecture and Methodology

The system engineering workflow followed an iterative path from conceptual sketching, Computer-Aided Design (CAD), finite sizing calculations, component fabrication, sensory electronics integration, to validation trials (Figure 1).

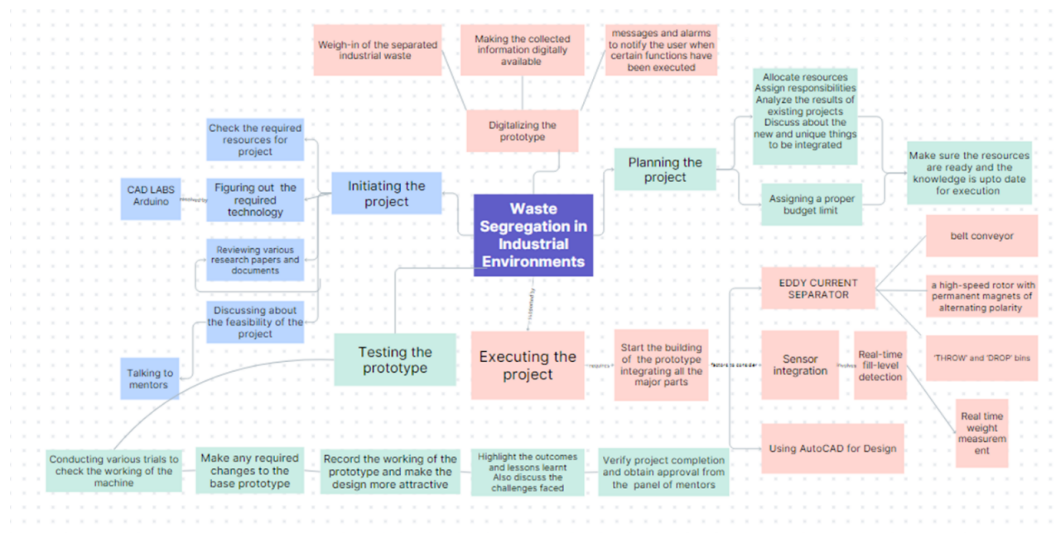


Figure 1: System architecture and project execution workflow spanning conceptualization, mechanical fabrication, sensor integration, and validation.

5.1 Design Phase

- **2D Mechanical Drafting:** Detailed part drawings and dimensional tolerances were created in AutoCAD to guide machine-shop turning and milling operations.
- **3D Parametric Modeling:** Complete 3D virtual assemblies were modeled in Autodesk Fusion 360 to verify bearing seat alignments, motor mounting clearances, and center-of-gravity balance.
- **Circuit Simulation:** Embedded sensor networks and signal conditioners were prototyped and verified in Autodesk TinkerCAD before breadboard soldering.

5.2 Component Procurement and Material Selection Rationale

Materials were selected based on structural rigidity, electromagnetic neutrality, wear resistance, and cost effectiveness:

- **Extruded Aluminum Profiles (30×30):** Chosen for high strength-to-weight ratio, structural modularity, and zero magnetic interference.
- **En-8 Carbon Steel (Zinc Coated):** Used for drive shafts and roller flanges to ensure high torsional shear strength and corrosion protection.
- **Nylon Outer Shell Flanges:** Non-conductive, low-dielectric polymer sleeves prevent eddy current self-heating within structural endplates.
- **Smooth PVC Pipe Drum Sleeve:** Acts as an electromagnetic barrier and physical shield, isolating the high-speed internal rotor from the slower outer conveyor belt.
- **PU-Coated Belt:** Durable, anti-static, heat-resistant conveyor belt ensuring stable material transport without erratic slip.

6 Engineering Tools and Techniques

6.1 Software Toolchain

1. **AutoCAD:** Used for 2D drafting of precision shafts, bracket mountings, and flange bolt patterns.
2. **Autodesk Fusion 360:** Used for 3D modeling, dynamic collision checks, and parametric rendering (Figure 2).

3. **Arduino IDE:** Used to write, compile, and flash embedded C++ firmware onto the microcontroller.
4. **Autodesk TinkerCAD:** Provided circuit schematics and breadboard validation for sensor amplifier logic.
5. **MIT App Inventor:** Visual blocks platform used to construct the responsive Android monitoring application.

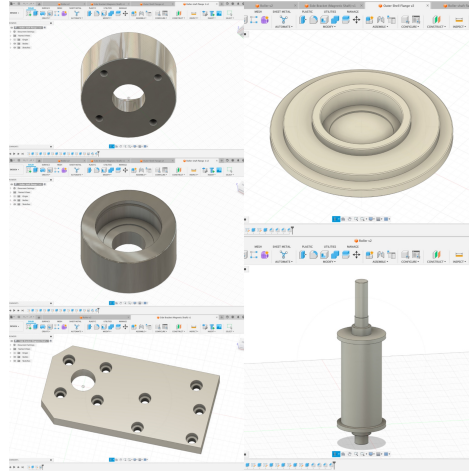
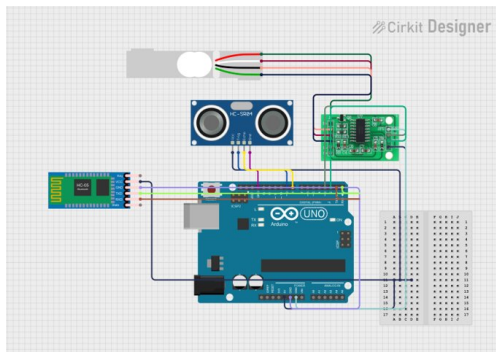


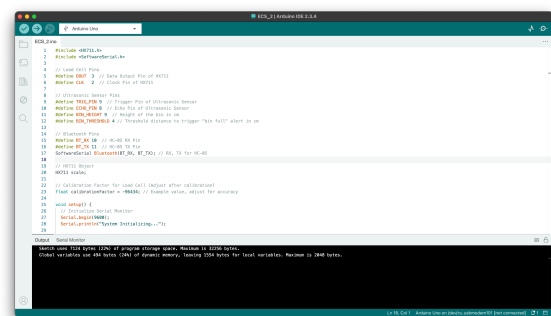
Figure 2: Autodesk Fusion 360 parametric 3D models and renders of custom-fabricated components (roller shaft flanges, nylon outer sleeves, and bearing brackets).

6.2 Hardware and Sensory Stack

- **Microcontroller (Arduino Uno):** ATmega328P core running at 16 MHz, orchestrating sensor sampling and serial packet streaming.
- **Load Cell (20 kg Parallel Beam):** Strain gauge transducer converting mechanical deflection into proportional differential millivolt signals.
- **HX711 Instrumentation Amplifier:** 24-bit high-precision analog-to-digital converter configured with an internal low-noise programmable gain amplifier (PGA gain = 128).
- **HC-SR04 Ultrasonic Sensor:** Emits 40 kHz ultrasonic pulses to measure the distance to the accumulated scrap pile in the bin.
- **HC-05 Bluetooth Module:** Serial Port Profile (SPP) transceiver transmitting weight and status strings at 9600 baud.



(a) TinkerCAD virtual circuit simulation.



(b) Arduino IDE firmware environment.

Figure 3: Embedded hardware simulation and software implementation environment.

7 Prototype Fabrication and Mechanical Assembly

7.1 Bill of Materials and Component Classification

Table 2 outlines the standard off-the-shelf, custom-machined, and miscellaneous parts incorporated into the physical build.

Table 2: Comprehensive Bill of Materials (BOM) and Component Classification

Category	Component	Material / Technical Specification
Off-the-Shelf Parts	Aluminum Profiles	30 × 30 mm modular extruded aluminum
	Deep Groove Ball Bearings	HCH IATF 16949:2016 certified bearings
	Circlips	High-carbon spring steel (Types A, B, and E)
	Industrial Belt	Static-dissipative, heat and tear resistant
	Fasteners	M6 Allen bolts with top-loading Slot-8 T-nuts
	Structural Brackets	30 × 30 mm Die-cast aluminum angle brackets
	Drive Sprockets & Pulleys Drive Chains	Industrial machined steel Precision linking roller chain
Custom Fabricated	Support Brackets	Machined 6082-T6 structural aluminum
	Roller Shaft Flange	En-8 high-strength steel (Zinc-plated)
	Idle Roller Flange	En-8 high-strength steel (Zinc-plated)
	Outer Shell Flange	CNC-turned wear-resistant Nylon
Drive & Assemblies	High-Speed Magnetic Rotor	Multi-pole NdFeB permanent magnets salvaged from servo rotor
	Drum Outer Sleeve	Schedule-40 non-magnetic PVC pipe
	Material Conveyor Belt	Heavy-duty polyurethane (PU) coated belt
	Safety Enclosures	Sheet metal chain and belt guards
	Speed Reduction Gearbox Three-Phase Induction Motors	15:1 ratio worm-reduction unit 2790 RPM rated industrial induction motors

7.2 Assembly Sequence

The physical separator was constructed and commissioned over an 18 man-hour assembly window across three days:

1. **Mainframe Construction:** Rigid 30 × 30 extruded aluminum sections were assembled using gussets and T-nuts, providing an adjustable modular base frame.
2. **Bearing Block & Bracket Installation:** Custom-machined aluminum side brackets were anchored to the frame, ensuring axial alignment across both primary shafts.

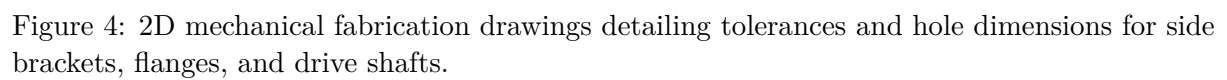


Figure 5: Initial conceptual framing schematic showing belt conveyor and roller placement.



Figure 6: Procured hardware components, aluminum extrusion framing, bearings, fasteners, motor, and machined mechanical parts.

3. **Magnetic Rotor Isolation Sleeve:** A non-conductive PVC cylinder was machined to house the inner high-speed magnetic rotor, mechanically isolating its rotation from the outer conveyor drum.
4. **Roller and Conveyor Belt Fitting:** Bearings were seated into the end flanges, the PU conveyor belt was looped over the pulleys, and tension was balanced to prevent tracking wander.
5. **Motor Flange Mounting:** Two AC motors were mounted via B14 flange plates directly onto the extrusion channels.
6. **Chain Drive Transmission:** Keyed sprockets were locked onto the motor and drum shafts; roller chains were installed with proper slack tension.
7. **VFD Electrical Integration:** A single-phase to three-phase Variable Frequency Drive (VFD, 1 HP) was wired in a star configuration to govern motor speeds accurately.
8. **Dynamic Alignment and Tuning:** Belt tracking was tuned, rotational vibration was dampened, and electrical interlocks were calibrated.

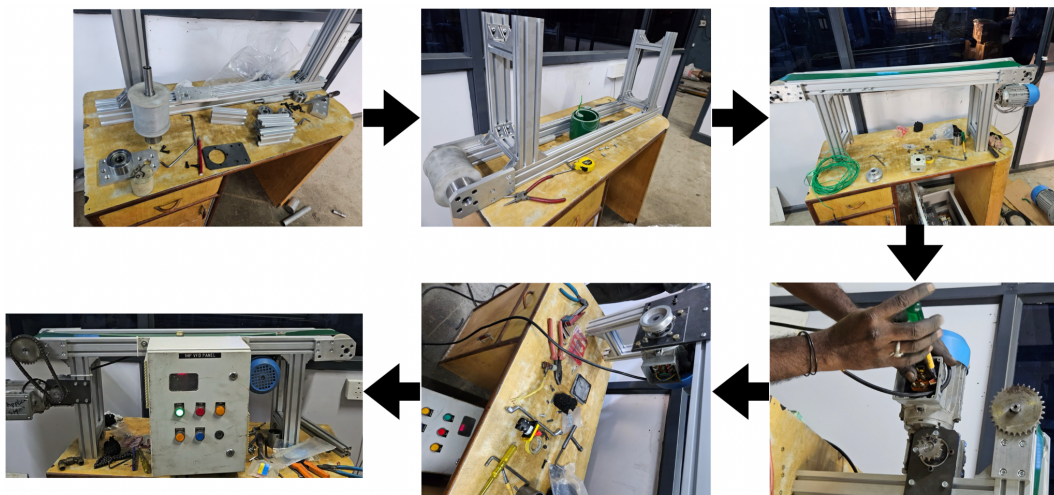


Figure 7: Step-by-step photographic record of frame erection, motor mounting, transmission fitting, and electrical cabinet wiring.

7.3 Engineering Challenges and Practical Resolutions

During fabrication and mechanical commissioning, several technical issues were resolved:

1. **Hardware Quantity Underestimation:** T-nuts and M6 socket cap screws were consumed rapidly; a centralized stock of fasteners was procured to avoid future stoppages.
2. **Missing B14 Motor Flange Adapter:** The specific motor flange was out of stock. Two steel plates were CNC milled and dowel-pinned together to produce a custom adapter plate.
3. **Conveyor Belt Tracking Sag:** Uneven tension caused belt wandering and friction against the side frame. Adjustable tensioning slots were machined into the idler roller brackets.
4. **Drive Motor Power Insufficiency:** The original 1478 RPM motor lacked sufficient centrifugal torque for high-speed rotor excitation. It was upgraded to a ~ 3000 RPM induction motor.
5. **VFD Single-to-Three Phase Wiring Complexity:** Industrial three-phase motors were adapted to single-phase utility mains by configuring stator windings in star configuration across a compact 1 HP inverter drive.
6. **Conveyor Linear Speed Mismatch:** A 7.5 : 1 gearbox ran the conveyor too fast, launching materials off the belt prematurely. Replacing it with a 15 : 1 reduction gearbox achieved optimal linear feed speed.
7. **Chassis Leveling and Axial Runout:** Misaligned bearing blocks consumed 2 hours of realignment. Dial indicators and precision spirit levels were employed to ensure coaxial rotation.
8. **Material Ejection Containment:** Scattered fragments occasionally bounced off the belt; transparent polycarbonate side skirts were affixed above the conveyor.
9. **High-Speed Operator Safety:** Rapidly rotating sprockets and chains represented pinch hazards; sheet metal chain guards and emergency stop pushbuttons were added.

8 Electronics, Sensing, and Supervisory App

8.1 Circuit Topology and Pinout

The electronic subsystem connects the sensing elements to the central Arduino Uno microcontroller, as mapped in Table 3 and illustrated in Figure 8.

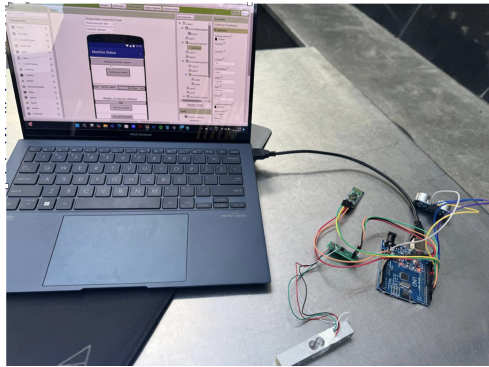
8.2 Load Cell Signal Conditioning and Calibration

The parallel beam load cell generates differential signals on the order of ± 20 mV. The HX711 integrates this signal with a 24-bit Σ - Δ ADC:

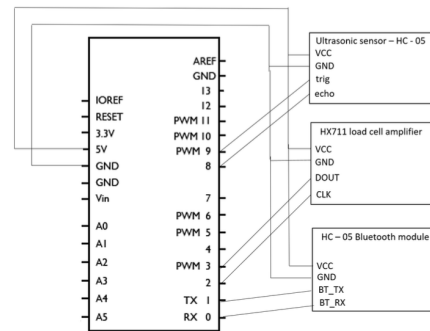
1. **Raw Zero-Load Offset:** The uncalibrated analog offset produced an initial raw ADC value of -7050 .
2. **Known Reference Calibration:** Applying calibrated test masses (1.000 kg and 2.500 kg) allowed computation of the linear slope.
3. **Final Calibration Factor:** The finalized scaling multiplier was programmed as -96343 , providing mass resolutions within ± 10 grams.

Table 3: Microcontroller Pin Connections and Peripheral Interfacing

Arduino Uno Pin	Peripheral Module	Module Pin	Functional Description
Pin 2	HX711 Amplifier	CLK	Serial Clock output for ADC conversion
Pin 3	HX711 Amplifier	DOUT	Serial Data line for 24-bit weight reading
Pin 8	HC-SR04 Ultrasonic	ECHO	Time-of-flight pulse echo input
Pin 9	HC-SR04 Ultrasonic	TRIG	10 μ s ultrasonic pulse trigger
Pin 10	HC-05 Bluetooth	TX	SoftwareSerial RX pin
Pin 11	HC-05 Bluetooth	RX	SoftwareSerial TX pin (via resistor divider)
5V	All Peripherals	VCC	Regulated 5V DC system power bus
GND	All Peripherals	GND	Common ground reference



(a) Schematic pinout diagram.



(b) Physical breadboard and sensor test setup.

Figure 8: Electronic interfacing between Arduino Uno, HX711, ultrasonic transducer, and Bluetooth transceiver.

8.3 Firmware Implementation

The core Arduino control routine samples the load cell, computes distance thresholds, and formats a serial telemetry packet every second:

```

1  #include <HX711.h>
2  #include <SoftwareSerial.h>
3
4  #define DOUT      3    // Data Output Pin of HX711
5  #define CLK       2    // Clock Pin of HX711
6  #define TRIG_PIN  9    // Ultrasonic Trigger Pin
7  #define ECHO_PIN  8    // Ultrasonic Echo Pin
8  #define BIN_THRESHOLD 4 // Distance threshold in cm for "bin full"
9
10 SoftwareSerial Bluetooth(10, 11); // RX, TX for HC-05
11 HX711 scale;
12 const float calibrationFactor = -96343.0;
13
14 void setup() {
15   Serial.begin(9600);
16   Bluetooth.begin(9600);
17   scale.begin(DOUT, CLK);
18   scale.tare();
19   scale.set_scale(calibrationFactor);
20   pinMode(TRIG_PIN, OUTPUT);

```

```

21   pinMode(ECHO_PIN, INPUT);
22 }
23
24 int getDistance() {
25   digitalWrite(TRIG_PIN, LOW);
26   delayMicroseconds(2);
27   digitalWrite(TRIG_PIN, HIGH);
28   delayMicroseconds(10);
29   digitalWrite(TRIG_PIN, LOW);
30   long duration = pulseIn(ECHO_PIN, HIGH);
31   return duration * 0.034 / 2; // Convert to cm
32 }
33
34 void loop() {
35   float weight = scale.get_units(10); // Average of 10 readings
36   int distance = getDistance();
37   bool binFull = (distance <= BIN_THRESHOLD);
38
39   // Transmit telemetry over Bluetooth SPP
40   Bluetooth.print("Weight: ");
41   Bluetooth.print(weight, 2);
42   Bluetooth.println(" kg");
43   Bluetooth.println(binFull ? "Bin Status: FULL" : "Bin Status: OK");
44
45   delay(1000);
46 }

```

Listing 1: Embedded Arduino firmware running on the ATmega328P controller.

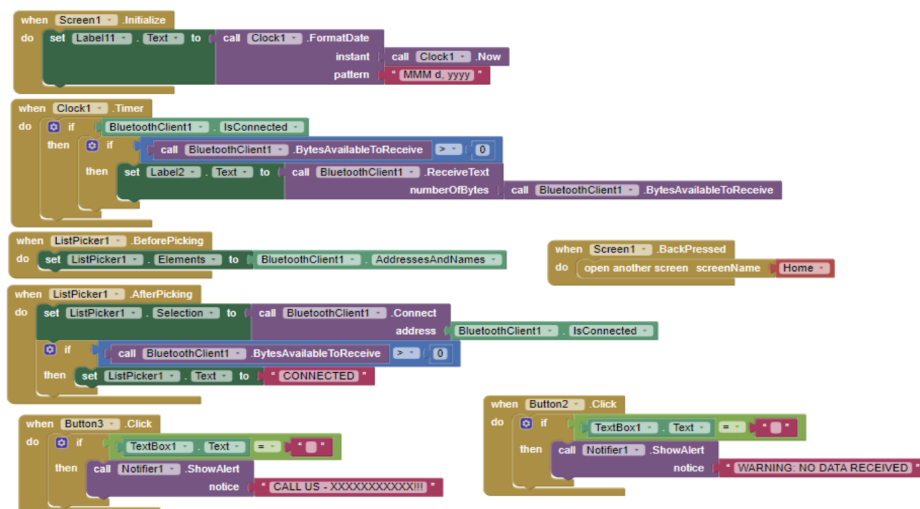


Figure 9: MIT App Inventor visual logic blocks handling Bluetooth SPP string parsing, timer events, and alert dialogs.

8.4 Supervisory Mobile Dashboard

The MIT App Inventor application was designed with automated reconnect routines to mitigate RF dropped packets. As displayed in Figure 10, the application transitioned from an early wireframe to a full supervisory telemetry interface featuring live weight display, distance-to-brim readouts, bin-full warnings, and cloud upload hooks.

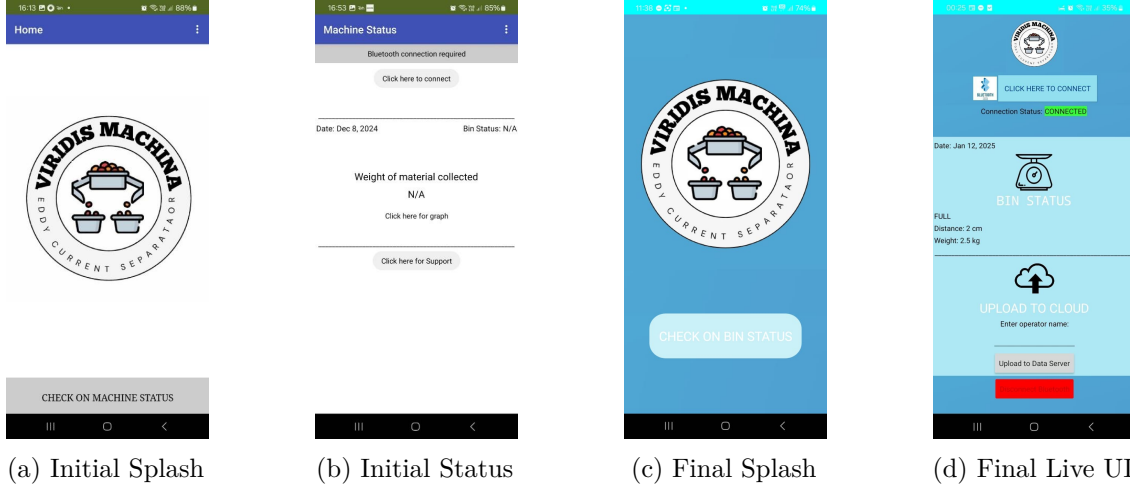


Figure 10: Evolution of the mobile monitoring app from early mockups to the final operational dashboard.

9 Experimental Results and Discussion

9.1 Non-Ferrous Separation Performance

Sorting experiments were carried out on uniform standardized test pieces (1 cm² surface area, 1 mm thickness) fed continuously along the conveyor belt. The results are summarized in Table 4.

Table 4: Deflection Test Results Across Non-Ferrous Metallic Specimens (1 cm² × 1 mm)

Material	Density (g/cm ³)	Conductivity (MS/m)	Measured Throw Distance (m)
Aluminum	2.70	~ 37.0	1.25 – 1.70
Copper	8.96	~ 58.0	1.00 – 1.60
Brass	~ 8.50	~ 15.0	0.90 – 1.50

9.2 Physical Dynamics and Lorentz Trajectory Analysis

The ejection trajectory of a non-ferrous particle in an ECS is governed by its **conductivity-to-density ratio** (σ/ρ):

- **Aluminum** ($\sigma/\rho \approx 13.7 \text{ k} \cdot \text{m}^2/\Omega \cdot \text{kg}$): Exhibits the highest throw distance (1.25–1.70 m) due to its low mass density relative to electrical conductivity. The magnetic repulsive force accelerates the particle clear of the discharge pulley.
- **Copper** ($\sigma/\rho \approx 6.5 \text{ k} \cdot \text{m}^2/\Omega \cdot \text{kg}$): Despite higher electrical conductivity ($\approx 58 \text{ MS/m}$), copper’s substantial density (8.96 g/cm³) increases gravitational pull, requiring greater repulsive Lorentz force. Thicker copper billets exhibited lower deflection.
- **Brass** ($\sigma/\rho \approx 1.8 \text{ k} \cdot \text{m}^2/\Omega \cdot \text{kg}$): Yielded the smallest throw distance (0.90–1.50 m), directly matching theoretical expectations due to lower alloy conductivity.

Geometry also influenced trajectory: planar flakes and hollow rings experienced higher induced eddy current loops and cleaner ejections compared to tightly coiled wires or thick ingots.

9.3 Telemetry and Weight Sensor Reliability

Continuous material drops onto the load-cell receiver bin demonstrated reliable linear response across 0.05 kg to 15.0 kg. The calibrated scale factor (−96343) maintained steady measurement stability without zero-drift during continuous 30-minute operational runs. Bluetooth packets

were transmitted and refreshed once per second without noticeable UI latency on the mobile terminal.

10 Conclusion and Future Scope

10.1 Conclusion

This work successfully developed, assembled, and experimentally validated a benchtop Eddy Current Separator with an integrated IoT telemetry subsystem. The mechanical design achieved clean separation across aluminum, copper, and brass test fragments. Real-time mass logging and container capacity tracking operated robustly over a wireless Bluetooth connection. The resulting machine provides a low-cost, modular blueprint for automated non-ferrous recovery in localized scrap reclamation contexts.

10.2 Future Scope

- **Automated Dynamic Splitter:** Incorporating servo-actuated diverter plates controlled by computer vision or optical speed sensors to adjust splitter angles dynamically for mixed feed streams.
- **Upstream Pre-Conditioning / Shredder Feed:** Integrating an in-line mechanical rotary shredder to achieve uniform particle size distribution prior to belt loading.
- **High-Flux Neodymium Rotors:** Increasing rotor pole density and utilizing N52 grade NdFeB magnets to enable effective separation of heavy copper fragments and smaller sub-millimeter fines.
- **Industrial IoT / MQTT Gateway:** Upgrading the microcontroller to an ESP32 or Raspberry Pi to stream data directly to cloud-based dashboards (e.g., Grafana / AWS IoT) over Wi-Fi.